Identifiability-Aware Source Apportionment in City-Scale Advection-Diffusion Systems

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Abstract

Source apportionment from sparse urban air-quality sensors is an inverse problem whose resolution is limited by sensor placement, wind-driven transport, background variation, and noise. Known or proxy emission inventories make attribution scientifically meaningful by restricting the unknown source field to a finite set of candidate source groups, but they do not guarantee that those groups are distinguishable from the available observations. We represent time-varying source activity with a low-dimensional nonnegative temporal basis and formulate inventory-based source apportionment as a wind-conditioned lagged inverse problem in which each source–basis coefficient produces a sensor-time fingerprint. After projecting out a separate low-dimensional background space, the relevant object is the projected lagged source–basis response matrix ($\widetilde{H}_\Phi$). We show that exact identifiability at the chosen temporal-basis resolution requires full column rank of $\widetilde{H}_\Phi$, while noise-robust attribution is controlled by its singular values, coefficient visibility, background absorption, pairwise fingerprint coherence, and ray distance. Building on this analysis, we propose an identifiability-aware apportionment (IASA) framework that estimates nonnegative source–basis coefficients, reconstructs source activity trajectories, and reports uncertainty and conservative deterministic grouping recommendations for indistinguishable sources. We instantiate the framework on a New Delhi platform built from government $PM_{2.5}$ and wind observations, regulatory sensor locations, and seven named proxy inventories, and define controlled and observed-data evaluations of recovery, ambiguity, wind diversity, background stress, transport error, inventory robustness, and residual model adequacy. The framework reports the attribution resolution defensible conditional on the declared inventories, transport, background, lag, and noise model rather than presenting the most detailed possible attribution vector.

1. Introduction

Urban air pollution is driven by local emissions, meteorological transport, regional background pollution, and sensor noise. For policy, the central question is often not only *where* pollution is high, but *which* source groups contributed to the observed concentrations. Source apportionment therefore seeks to estimate the contributions of categories such as traffic, industry, brick kilns, and population-related activity from a combination of source inventories, transport models, and concentration measurements.

This task is especially difficult in sparse sensor networks. A model may fit sensor trajectories accurately while still failing to identify the underlying source contributions. The reason is geometric: after wind transport and sparse observation, two distinct source groups can induce nearly identical sensor-time signatures. A strong distant source and a weaker nearby source, for example, may become observationally equivalent once transport attenuation, finite travel time, background correction, and noise are taken into account. In such cases, fine-grained source attribution is not merely statistically uncertain; it is not identifiable from the available sensing system.

We study this issue in the inventory-based setting. Instead of attempting to recover an arbitrary unknown source field, we assume access to a finite set of known or proxy source maps. These inventories restrict attribution to a lower-dimensional source model and make the problem interpretable. To allow realistic time variation without estimating one free value at every hour, we represent each source activity as a nonnegative combination of a small number of temporal basis functions. The unknowns are therefore source–basis coefficients, from which source activity trajectories and contributions are reconstructed. However, the inventory and temporal-basis restrictions alone do not make the problem identifiable. Coefficient fingerprints can still be weakly visible, absorbed by

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background components, or mutually coherent after transport to the sensor network.

The key object in our analysis is a projected lagged source-response matrix. For a fixed wind-field realization, transport operator, source inventory, sensor layout, and lag window, each source–basis pair induces a finite-horizon sensor-space fingerprint. Stacking these fingerprints gives a lagged source–basis response matrix H_Φ^{lag}. Real observations also contain background and temporal components, represented by a low-dimensional basis Q. After projecting out this background space, the inverse problem is governed by

$$\widetilde{H}_\Phi = P_Q^\perp H_\Phi^{\mathrm{lag}},$$

where $P_Q^\perp$ is the projection onto the orthogonal complement of the background basis. The rank, singular values, coefficient visibility, background absorption, pairwise coherence, and ray distance of $\widetilde{H}_\Phi$ determine which temporal source contributions can be separated and which should be reported only after merging or qualification. A constant-activity model which measures the average activity over the observation period is recovered by using one constant basis function per source.

This perspective changes the role of source apportionment models. The goal is not to return the most detailed vector of source contributions permitted by the inventory. It is to report a conservative source resolution that is defensible under the available sensors, wind regime, background correction, and noise level, conditional on the completeness and correctness of the declared source model. An apportionment method should therefore output source estimates alongside identifiability diagnostics and ambiguity certificates.

We make the following contributions:

- We formulate sparse-sensor, inventory-based source apportionment as a lagged wind-conditioned inverse problem over nonnegative source–temporal- basis coefficients.
- We identify the projected lagged response matrix $\widetilde{H}_\Phi = P_Q^\perp H_\Phi^{\mathrm{lag}}$ as the central object governing coefficient separability after transport, sparse observation, lag, and background correction.
- We derive diagnostics for exact and noise-robust identifiability, including rank, smallest singular value, condition number, effective rank, coefficient visibility, background absorption, pairwise fingerprint coherence, and ray distance.
- We propose an identifiability-aware apportionment procedure that fits nonnegative source–basis coefficients, reconstructs source activity trajectories, reports uncertainty, flags weakly visible coefficients, and recommends connected report groups when eligible coefficient fingerprints are indistinguishable under the current sensing system.
- We instantiate the framework on a New Delhi platform comprising government $\mathrm{PM}_{2.5}$ and wind records, regulatory sensor locations, FieldFormer-based gridded wind imputation, and seven named normalized proxy inventories.
- We define controlled and observed-data evaluations of conditioning, source ambiguity, background stress, wind diversity, temporal-basis recovery, response error, uncertainty, and residual structure.

The broader message is that sparse-sensor source apportionment is limited by the information content of the sensing system. Reliable attribution requires not only a transport model and an optimizer, but an explicit account of which source–basis fingerprints are visible and separable in the observed sensor space.

2. Related Work

Air-pollution source apportionment. Source apportionment has a long history in air-quality science, with receptor-oriented methods estimating source contributions from ambient chemical or physical measurements and source-oriented methods propagating emissions inventories through atmospheric transport models. Classical receptor frameworks include chemical mass balance and factor-analytic approaches such as positive matrix factorization (Watson et al., 2002; Paatero & Tapper, 1994; Hopke, 2016). Reviews and harmonization efforts have clarified best practices for receptor models, source-oriented models, and their combined use, while also documenting practical sensitivity to source profiles, factor interpretation, chemical resolution, and model comparability (Viana et al., 2008; Belis et al., 2013; 2019; Mircea et al., 2020; Hopke et al., 2020). These sensitivities arise because ambient measurements provide only indirect mixtures of source contributions: source profiles may be incomplete or locally unavailable, tracers and proxy inventories can be collinear, factor decompositions may admit multiple plausible interpretations, and transport or inventory assumptions can map distinct source configurations to similar observations. This ambiguity is especially visible in settings such as India, where studies can produce divergent source-contribution estimates for the same city because of limited local profiles, collinear tracers, and methodological differences (Pant & Harrison, 2012); global reviews further show that policy-relevant source categories such as traffic, industry, dust, and domestic combustion are routinely reported from heterogeneous apportionment studies (Karagulian et al., 2015). Our work is complementary to this literature: rather than proposing another receptor model or emissions inventory, we analyze when known or proxy source maps are distinguishable from sparse concentration sensors after wind-conditioned transport, finite lags, background correction, and noise.

Inverse modeling and physics-constrained learning. Estimating latent states or parameters from partial observations is central to inverse problems and data assimilation (Tarantola, 2005; Engl et al., 1996). Classical approaches such as Kalman filtering, ensemble Kalman filters, and variational data assimilation use dynamical models together with priors, covariance assumptions, and observation models to infer hidden quantities from measurements (Wang et al., 2023; Carrassi et al., 2018). More recent physics-informed and operator-learning methods, including PINNs and neural operators, incorporate governing equations or physics-motivated structure into flexible learning systems for parameter estimation and system identification (Raissi et al., 2019; Li et al., 2024; 2020; Bhardwaj et al., 2025b). These methods motivate our use of a transport-structured, PyTorch-based constrained fitting backend, but they do not by themselves resolve the identifiability question. In sparse sensing regimes, the map from latent source activities, transport parameters, and background components to sensor observations can be many-to-one: different parameterizations may fit the same measurements after optimization. Our focus is therefore complementary to calibration and physics-constrained learning methods. Rather than asking only how to fit the projected sensor data, we characterize when the fitted source–basis coefficients are identifiable from the projected lagged response matrix and when the result should be weakened, qualified, or grouped.

Sensor-space spatio-temporal learning. Graph neural networks, transformers, and sequence models have been widely used to model spatio-temporal processes from sensor data, including traffic, environmental, and air-quality time series (Li et al., 2017; Yu et al., 2017; Wu et al., 2019; Chen et al., 2023; Iyer et al., 2022; Bhardwaj et al., 2025a;b). These methods are relevant to our setting because they operate in the same sparse-sensor observational regime. However, their usual objective is forecasting, interpolation, or representation learning over sensor measurements rather than attribution to physically meaningful source inventories. A sensor-space model can predict observed concentrations well while leaving unresolved whether distinct source groups would produce distinguishable wind-conditioned fingerprints at the deployed sensors. Our work is therefore complementary: instead of learning an unconstrained predictive representation of the sensor field, we analyze the identifiability of declared source–basis contributions under a specified transport, background, lag, and noise model.

Observability and system identification. The question of whether latent quantities can be recovered from observations is closely related to classical notions of observability in control theory (Chen & Chen, 1984). For linear dynamical systems, observability characterizes whether the internal state can be uniquely determined from output trajectories. Extensions to nonlinear systems and PDEs have been studied in system identification and inverse-problem literature (Ljung et al., 1987; Banks & Kunisch, 2012; Colton et al., 1990), often under assumptions of sufficient measurement richness. We bring this perspective to inventory-based air-pollution source apportionment under sparse spatial sensing. Rather than asking whether a full latent state or physical model is observable, we ask whether declared source–basis contributions have distinct projected sensor-time fingerprints after wind-conditioned transport, finite lags, background correction, and noise. This leads to diagnostics tailored to the apportionment task, including rank and singular-value stability, source visibility, pairwise coherence, background absorption, and conservative grouping, together with a real-world instantiation that reports the attribution resolution supported by the available sensors, inventories, and transport assumptions.

3. Problem Setup: Sparse Sensors, Wind, and Source Apportionment

We formalize source apportionment as a finite-dimensional inverse problem in which known source inventories are transported to a sparse sensor network under a wind-conditioned lagged operator. This section defines the observation model, the inventory parameterization, the background correction, and the projected response matrix used throughout the paper.

City-scale transport model. Let $u(\mathbf{x}, t) \in \mathbb{R}$ denote the pollutant concentration field on $\Omega \subset \mathbb{R}^d$. A generic continuous-time model is

$$\frac{\partial u}{\partial t} = \mathcal{F}_\eta(u, \mathbf{x}, t; w(t)) + s(\mathbf{x}, t) + b(\mathbf{x}, t), \qquad (1)$$

where $w(t)$ is the meteorological state, s is local source emission, b is background or regional pollution, and η denotes transport and dispersion parameters. The exact dynamics may be approximated by a plume model, an advection–diffusion solver, or a learned surrogate constrained by transport structure.

Sparse observations. Discretize the concentration field on n grid cells, so that $\mathbf{u}_t \in \mathbb{R}^n$. Let $X_{\text{sens}} = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ be the fixed sensor locations with $m \ll n$. The observation operator $O : \mathbb{R}^n \to \mathbb{R}^m$ samples or interpolates the latent field at these locations:

$$\mathbf{y}_t = O\mathbf{u}_t + \boldsymbol{\epsilon}_t, \qquad \mathbf{y}_t \in \mathbb{R}^m. \qquad (2)$$

Stacking over $t = 1, \ldots, T$ gives

$$Y = [\mathbf{y}_1^\top, \ldots, \mathbf{y}_T^\top]^\top \in \mathbb{R}^{mT}. \qquad (3)$$

Because m is small relative to the field dimension, many latent fields and source configurations can be consistent with the same sensor trajectory.

Inventory-based source parameterization. We assume a set of K known or proxy source maps,

$$S = [\mathbf{s}_1\ \mathbf{s}_2\ \cdots\ \mathbf{s}_K] \in \mathbb{R}^{q\times K}, \tag{4}$$

where $\mathbf{s}_k \in \mathbb{R}^q$ is the spatial pattern of source group k. The source grid dimension q may differ from the concentration-grid dimension n. We represent time-varying source activity with a fixed, predeclared nonnegative temporal dictionary

$$\Phi = [\boldsymbol{\phi}_1\ \cdots\ \boldsymbol{\phi}_B] \in \mathbb{R}_+^{T\times B}, \qquad C = [c_{kb}] \in \mathbb{R}_+^{K\times B}. \tag{5}$$

The activity of source k at time t is

$$\theta_k(t) = \sum_{b=1}^{B} c_{kb}\phi_b(t), \qquad \theta_k(t) \geq 0. \tag{6}$$

The basis Φ is shared as a dictionary of admissible temporal patterns, not as a claim that all sources follow the same activity trajectory: each source has its own nonnegative mixture weights c_{kb}. In applications, these basis functions may encode predeclared calendar, diurnal, seasonal, regulatory, or inventory-derived activity patterns, and source-specific prior knowledge can be imposed by masking inadmissible source–basis coefficients. We do not learn Φ from the same sparse concentration observations used for apportionment; jointly estimating both Φ and C would introduce additional bilinear non-identifiability beyond the source distinguishability analyzed here. The activity bases used for source emissions are nonnegative; signed temporal harmonics, when used, belong to the separate background basis Q. We vectorize C in source-major, basis-minor order as $\mathbf{c} = \mathrm{vec}_{\mathrm{src}}(C) \in \mathbb{R}_+^J$, where $J = KB$. The constant source activity model is the special case $B = 1$, $\phi_1(t) = 1$, and $c_{k1} = \theta_k$.

Lagged wind-conditioned response. Emissions released at time $t-\ell$ may affect sensors at time t only after being transported by the intervening wind field. Let

$$G^{\partial}_{t,t-\ell}(w_{t-\ell:t}) \in \mathbb{R}^{n\times q} \tag{7}$$

map emissions released on the source grid at time $t-\ell$ to the concentration grid at time t. The superscript ∂ denotes an open-boundary operator: mass transported outside Ω leaves the modeled domain rather than being reflected or wrapped around. For a maximum lag L, define

$$\mathcal{L}_t = \{0, 1, \ldots, \min(L, t-1)\}. \tag{8}$$

For source k and basis component b, the corresponding unit-coefficient sensor fingerprint at time t is

$$\mathbf{h}^{\mathrm{lag}}_{t,kb} = \sum_{\ell\in\mathcal{L}_t} \phi_b(t-\ell) O G^{\partial}_{t,t-\ell}(w_{t-\ell:t})\mathbf{s}_k \in \mathbb{R}^m. \tag{9}$$

Stacking over time and arranging columns in source-major, basis-minor order gives

$$\begin{aligned} H^{\mathrm{lag}}_{\Phi} &= [\mathbf{h}^{\mathrm{lag}}_{11}\ \cdots\ \mathbf{h}^{\mathrm{lag}}_{1B}\ \cdots\ \mathbf{h}^{\mathrm{lag}}_{KB}] \in \mathbb{R}^{mT\times J}, \\ J &= KB, \end{aligned} \tag{10}$$

where each $\mathbf{h}^{\mathrm{lag}}_{kb} \in \mathbb{R}^{mT}$ is the finite-horizon unit-coefficient sensor fingerprint, which is the sensor-time pattern produced by one unit of coefficient c_{kb}. When $B = 1$, these columns reduce to the fingerprints of the constant-activity model.

The lag window changes these fingerprints and the release times contributing to each observation; it does not add sensor–time observations or change the row count. We choose it before fitting source coefficients. Physical travel-time and residence-time bounds define an increasing candidate grid $\mathcal{L}_{\mathrm{cand}}$. For adjacent candidates L and $L+\Delta$, with the same observed rows and columns, define

$$\eta_L = \frac{\|H^{\mathrm{lag}}_{\Phi}(L+\Delta) - H^{\mathrm{lag}}_{\Phi}(L)\|_F}{\max\{\|H^{\mathrm{lag}}_{\Phi}(L+\Delta)\|_F, \epsilon\}}, \tag{11}$$

where the subscript F refers to the Frobenius norm: $\|A\|_F = \left(\sum_{i,j} A_{ij}^2\right)^{1/2}$. The primary lag is the smallest candidate with $\eta_L \leq \tau_L$, using $\tau_L = 10^{-3}$ by default. Rank, conditioning, and ambiguity components are also reported across the candidate grid. The fitted value of $\mathbf{c}$ is never used to select L.

Background and temporal correction. Observed concentrations include components not explained by the local source inventory, including regional background, smooth temporal trends, and sensor-specific offsets. We represent these components with a low-dimensional basis

$$Q \in \mathbb{R}^{mT\times r}, \qquad \boldsymbol{\gamma} \in \mathbb{R}^r, \tag{12}$$

and write

$$Y = H^{\mathrm{lag}}_{\Phi}\mathbf{c} + Q\boldsymbol{\gamma} + E. \tag{13}$$

The low-dimensionality ensures that source apportionment is not confounded by background correction.

Real concentration records need not be complete. Let $\mathcal{O}$ be the ordered set of observed sensor–time $PM_{2.5}$ rows and let $M_{\mathcal{O}} \in \{0,1\}^{N\times mT}$, $N = |\mathcal{O}|$, select them. The observation mask is applied identically to every row-aligned object:

$$Y_{\mathcal{O}} = M_{\mathcal{O}}Y, \quad H^{\mathrm{lag}}_{\Phi,\mathcal{O}} = M_{\mathcal{O}}H^{\mathrm{lag}}_{\Phi}, \quad Q_{\mathcal{O}} = M_{\mathcal{O}}Q. \tag{14}$$

Row metadata are filtered by the same ordered index set; an alignment mismatch is an error rather than an invitation to reorder silently. Wind may be imputed, but missing $PM_{2.5}$ values are not. For complete controlled observations, $M_{\mathcal{O}} = I$ and $N = mT$. Below we drop the $\mathcal{O}$ subscript and use $Y \in \mathbb{R}^N$, $H^{\mathrm{lag}}_{\Phi} \in \mathbb{R}^{N\times J}$, and $Q \in \mathbb{R}^{N\times r}$ for these aligned arrays.

Let

$$P_Q^{\perp} = I - QQ^{\dagger} \tag{15}$$

be the projection onto the orthogonal complement of the background space, with $Q^{\dagger}$ denoting the Moore–Penrose pseudoinverse. The corrected inverse problem is

$$\widetilde{Y} = \widetilde{H}_\Phi \mathbf{c} + \widetilde{E}, \qquad \widetilde{Y} = P_Q^\perp Y, \qquad \widetilde{H}_\Phi = P_Q^\perp H_\Phi^{\mathrm{lag}}. \tag{16}$$

All identifiability diagnostics in this paper are computed from $\widetilde{H}_\Phi$, not from the raw inventory matrix or the unprojected response.

Projected source–basis fingerprints. Writing

$$H_\Phi^{\mathrm{lag}} = [\mathbf{h}_{11}^{\mathrm{lag}} \ \cdots \ \mathbf{h}_{KB}^{\mathrm{lag}}], \qquad \widetilde{H}_\Phi = [\widetilde{\mathbf{h}}_{11} \ \cdots \ \widetilde{\mathbf{h}}_{KB}], \tag{17}$$

we have $\widetilde{\mathbf{h}}_{kb} = P_Q^\perp \mathbf{h}_{kb}^{\mathrm{lag}}$. These projected fingerprints encode the observable effect of each temporal component of each source after wind transport, lag, sparse sensing, and background correction. Source apportionment at the chosen temporal-basis resolution is well posed only to the extent that these coefficient fingerprints are visible and separable.

4. Wind-Conditioned Response Construction and Estimation

Section 3 defines the projected inverse problem. This section describes how the response matrix is constructed in practice and how source activities are estimated once the response has been formed.

4.1. Masked Wind Preparation

Government wind direction data is meteorological: it records where wind comes from, whereas transport requires the direction in which pollution moves. We first convert direction WD in degrees and speed WS to transport vectors,

$$\begin{aligned} U_x &= -WS\sin(WD\pi/180), \\ V_y &= -WS\cos(WD\pi/180). \end{aligned} \tag{18}$$

The observed station vectors and masks are supplied to a FieldFormer wind imputer (Bhardwaj et al., 2025b). FieldFormer is trained as a coordinate-query model: given sparse station observations $\{(\mathbf{z}_i, t, \mathbf{u}_{i,t}, m_{i,t})\}$, it predicts transport vectors at arbitrary spatial query locations and times. For the IASA response operator, we query the trained model on every response-grid cell $\mathbf{x}_g$ and hour t, producing

$$\widehat{\mathbf{w}}_t(\mathbf{x}_g) = \widehat{f}_\omega(\mathbf{x}_g, t; \{(\mathbf{z}_i, t', \mathbf{u}_{i,t'}, m_{i,t'})\}), \quad g = 1, \ldots, n. \tag{19}$$

This yields a gridded wind field $\widehat{W} \in \mathbb{R}^{T\times n\times 2}$, rather than a station-only or city-averaged sequence. Real-data validation is performed by masking observed station vectors and measuring held-out station-time error; dense city-wide wind truth is not assumed.

The imputed gridded wind field is an estimated transport input, not ground truth. To propagate wind-imputation uncertainty, we construct wind-field ensembles $\{\widehat{\mathbf{w}}_t^{(r)}(\mathbf{x}_g)\}_{r=1}^R$ using held-out-calibrated prediction error, station bootstrap resampling, checkpoint ensembles, or perturbations matched to validation residuals. Each ensemble member is queried on the same response grid and converted to grid displacement before response construction. The resulting response matrices are tagged as transport ensembles and are kept separate from inventory-robustness scenarios.

Meteorological speed must be converted to grid displacement before puff advection. For response timestep Δt_{s} seconds and cell dimensions $\Delta x_{\mathrm{m}}, \Delta y_{\mathrm{m}}$, we use

$$v_x^{\mathrm{grid}}(\mathbf{x}_g, t) = \frac{\widehat{U}_x(\mathbf{x}_g, t)\Delta t_{\mathrm{s}}}{\Delta x_{\mathrm{m}}}, \quad v_y^{\mathrm{grid}}(\mathbf{x}_g, t) = \frac{\widehat{V}_y(\mathbf{x}_g, t)\Delta t_{\mathrm{s}}}{\Delta y_{\mathrm{m}}}. \tag{20}$$

Every response artifact records these scales, the coordinate convention, the wind units, and the exact converted sequence. Controlled experiments use the same interface for constant, single-direction, diurnal, AR(1), and multi-directional synthetic winds.

4.2. Open-Boundary Lagged Plume Response

The transport response $G_{t,\tau}^{\partial}$ maps emissions released at $\tau \le t$ to the concentration field at time t. For any nonnegative emission vector $\mathbf{z} \in \mathbb{R}_+^q$, the open-boundary response obeys

$$G_{t,\tau}^{\partial}\mathbf{z} \ge 0, \qquad \mathbf{1}^\top G_{t,\tau}^{\partial}\mathbf{z} \le \mathbf{1}^\top \mathbf{z}. \tag{21}$$

Thus the operator preserves nonnegativity and is mass nonincreasing inside the modeled domain.

We instantiate this response with a Gaussian puff approximation. For a unit release from source cell i at location $\mathbf{r}_i$ and release time τ, the puff center is initialized as $\mathbf{z}_i(\tau) = \mathbf{r}_i$ and advected by the interpolated wind field,

$$\frac{d\mathbf{z}_i(a)}{da} = \widehat{\mathbf{w}}_a(\mathbf{z}_i(a)), \qquad a \in [\tau, t]. \tag{22}$$

Source cells have integer centers and domain extents $[-0.5, N_x - 0.5] \times [-0.5, N_y - 0.5]$. With substep size δa, the trajectory is discretized as

$$\mathbf{z}_i^{r+1} = \mathbf{z}_i^r + \delta a\, \widehat{\mathbf{w}}_{a_r}(\mathbf{z}_i^r). \tag{23}$$

The final substep is shortened so that the trajectory lands exactly at the requested observation time. If the center exits Ω, the whole remaining puff is removed and never reflected, wrapped, clamped, or reinserted. Otherwise its contribution is spread with an anisotropic Gaussian kernel

$$K_\psi(\mathbf{x}; \mathbf{z}_i, \Sigma_i) = \frac{\exp\left(-\frac{1}{2}\|\mathbf{x} - \mathbf{z}_i\|_{\Sigma_i^{-1}}^2\right)}{2\pi|\Sigma_i|^{1/2}}, \tag{24}$$

where ψ denotes dispersion parameters. A simple covariance model is

$$\Sigma_i(t,\tau) = R_i(t,\tau)\begin{bmatrix} \sigma_\parallel^2(t-\tau) & 0 \\ 0 & \sigma_\perp^2(t-\tau) \end{bmatrix} R_i(t,\tau)^\top, \tag{25}$$

with R_i aligned to the mean wind direction along the trajectory. When the mean wind norm is below the numerical threshold, the downwind direction is undefined. The implementation uses the positive-x axis as a deterministic tie-breaker for the anisotropic covariance orientation; this convention is recorded in the response metadata and affects only near-calm releases. The age in this covariance is lower-bounded by a positive minimum dispersion time, so same-time releases remain finite: effective age $= \max(t-\tau, t_{\min})$.

Response-matrix entries are computed by evaluating the kernel directly at sensor coordinates:

$$\left[OG^{\partial}_{t,\tau}(\widehat{\mathbf{w}}_{\tau:t};\psi)\mathbf{e}_i\right]_s = \chi_i(t,\tau)K_\psi(\mathbf{x}_s;\mathbf{z}_i(t),\Sigma_i(t,\tau)). \tag{26}$$

Here $\chi_i(t,\tau)$ is an open-boundary survival indicator for a release from source cell i at release time τ: it is one while the advected kernel center remains inside the modeled domain and zero after the release has exited. Boundary and truncation losses are recorded only as diagnostic quantities and are never used to renormalize the sensor response. Boundary loss denotes the portion of the Gaussian kernel lying outside the modeled spatial domain, while truncation loss denotes the portion omitted by evaluating retained mass only over a finite kernel support; both are estimated separately by grid quadrature over the response grid. These kernel losses are distinct from whole-release exit loss, where the advected release center leaves the open domain and no later sensor contribution is assigned to that release. Because the response matrix evaluates concentration at sensor points rather than integrating over area-weighted grid cells, summing entries across sensors or repeated sensor-time observations does not estimate retained pollutant mass.

Response matrices are constructed for source-induced increments relative to a declared initial-condition baseline. The baseline policy is fixed before source fitting, recorded with the response artifact, and applied consistently to the observations and response rows. Application-specific choices, such as whether to subtract a zero-source transported initial field or to use a zero-initial-state response, are part of the experimental instantiation rather than the generic response definition.

The open-boundary lagged plume response is implemented in PyTorch. Tensor operations retained in the graph can be differentiated, but discrete exit events and reporting logic are not assigned gradients.

4.3. Background Construction and Projection

The normal background basis may contain a global constant, centered and scaled elapsed-time polynomials, local-clock daily sine/cosine harmonics, reference-coded day effects, standardized regional sensor-coordinate trends, reference-coded sensor offsets, and a row-aligned user basis. It depends only on timestamps, day labels, sensor identity, and sensor coordinates. Its effective rank is capped at eight. Columns constructed from inventories, response fingerprints, or fitted source signals are excluded from the normal background basis; they are permitted only in explicitly labeled stress tests that measure how source-like background variation can absorb attribution signal.

The primary background basis Q is specified before source fitting: its columns and rank cap are chosen from the allowed source-independent metadata above, not by searching over Y or by optimizing source recovery. Once Q is fixed, its coefficients are estimated implicitly by projection, or explicitly in the joint fit. For the New Delhi study, the primary basis is the rank-four member of this allowed class: a global constant, centered linear trend, and first daily sine and cosine. Alternative source-independent bases are predeclared sensitivity analyses, not data-selected replacements for the authoritative model. Each comparison reports the resulting visibility, absorption, rank, conditioning, and ambiguity components.

Let $Q = U\Sigma V^\top$ be a thin singular value decomposition and retain columns U_r above the recorded rank tolerance. Projection is applied implicitly,

$$P_Q^\perp X = X - U_r(U_r^\top X), \tag{27}$$

without constructing an $N \times N$ matrix. The row metadata for Y, H_Φ^{lag}, and Q must be identical: each row must refer to the same timestamp, sensor identity, and sensor index in the same order before projection is applied.

4.4. Response Matrix and Source Activity Fit

Given the estimated wind field, source inventory, and temporal activity basis, the lagged unit coefficient source–basis response is constructed as in Section 3. Its (k,b)-th column is

$$\mathbf{h}^{\text{lag}}_{t,kb} = \sum_{\ell\in\mathcal{L}_t} \phi_b(t-\ell)OG^{\partial}_{t,t-\ell}(\widehat{\mathbf{w}}_{t-\ell:t};\psi)\mathbf{s}_k, \tag{28}$$

and stacking over time and source-major, basis-minor columns yields $H_\Phi^{\text{lag}} \in \mathbb{R}^{N\times J}$, $J = KB$, after applying the observation selection from Equation 14. Background correction gives

$$\widetilde{Y} = P_Q^\perp Y, \qquad \widetilde{H}_\Phi = P_Q^\perp H_\Phi^{\text{lag}}. \tag{29}$$

Scientific knowledge may declare a set $\mathcal{F}_0$ of coefficients to be identically zero before the response is diagnosed or fitted. With $\mathcal{I} = \{1,\ldots,J\}\setminus\mathcal{F}_0$, IASA removes the corresponding columns, computes every diagnostic from $\widetilde{H}_{\Phi,\mathcal{I}}$, fits only $\mathbf{c}_\mathcal{I}$, and then restores zeros in the full source–basis index. The default is $\mathcal{F}_0 = \varnothing$. Coefficients are not removed after fitting merely because their estimates are small; doing so would change the diagnosed inverse problem after seeing the data and could make the reported identifiability artificially favorable. Source–basis coefficients are then estimated by nonnegative regularized least squares,

$$\widehat{\mathbf{c}} = \arg\min_{\mathbf{c}\geq 0} \|\widetilde{Y} - \widetilde{H}_\Phi \mathbf{c}\|_2^2 + \lambda R(\mathbf{c}). \tag{30}$$

Reshaping $\widehat{\mathbf{c}}$ into $\widehat{C}$ reconstructs the source activity trajectory

$$\widehat{\theta}_k(t) = \sum_{b=1}^{B} \widehat{c}_{kb}\phi_b(t). \tag{31}$$

The regularizer can be ridge stabilization,

$$R(\mathbf{c}) = \|\mathbf{c}\|_2^2, \tag{32}$$

or a weak prior around inventory-derived activity levels,

$$R(\mathbf{c}) = \|\mathbf{c} - \mathbf{c}_0\|_2^2. \tag{33}$$

Equivalently, one may fit background and source coefficients jointly:

$$(\widehat{\mathbf{c}}, \widehat{\boldsymbol{\gamma}}) = \arg\min_{\mathbf{c}\geq 0, \boldsymbol{\gamma}} \|Y - H_\Phi^{\mathrm{lag}}\mathbf{c} - Q\boldsymbol{\gamma}\|_2^2 + \lambda R(\mathbf{c}). \tag{34}$$

The projected formulation is used for identifiability because it exposes the part of each source–basis fingerprint that remains after background correction.

4.5. Constrained End-to-End Refinement

IASA first constructs a fixed response matrix from the pretrained wind field and default dispersion parameters, fits source activities, and computes identifiability diagnostics for that declared response. It then performs a constrained local refinement of the wind field, dispersion parameters, source coefficients, and background coefficients. The refinement is initialized at the fixed-response solution and is allowed only to make small physically constrained corrections, so that improved sensor fit does not come from arbitrary changes to transport.

Let ϕ denote the wind-imputer parameters and let ψ denote the dispersion parameters of the puff response, such as $\sigma_\parallel$, $\sigma_\perp$, and the minimum dispersion age. Let ϕ_0 and ψ_0 be their pretrained or default values. We use R_{sm} for a predeclared wind-field smoothness penalty, such as temporal or spatial squared differences, and Ψ_{phys} for the physically admissible dispersion-parameter set. The refinement objective is

$$\begin{aligned}\mathcal{L}_{\mathrm{refine}} =& \|Y - H_\Phi^{\mathrm{lag}}(\phi,\psi)\mathbf{c} - Q\boldsymbol{\gamma}\|_2^2 + \lambda_\theta R(\mathbf{c}) \\ &+ \lambda_w \|\widehat{\mathbf{w}}_{1:T}^{\phi} - \widehat{\mathbf{w}}_{1:T}^{\phi_0}\|_2^2 + \lambda_\psi \|\psi - \psi_0\|_2^2 \\ &+ \lambda_{\mathrm{sm}} R_{\mathrm{sm}}(\widehat{\mathbf{w}}^{\phi}),\end{aligned} \tag{35}$$

subject to

$$\mathbf{c} \geq 0, \quad \psi \in \Psi_{\mathrm{phys}}, \quad \|\widehat{\mathbf{w}}_{1:T}^{\phi} - \widehat{\mathbf{w}}_{1:T}^{\phi_0}\|_\infty \leq \epsilon_w. \tag{36}$$

The refined response is accepted only as a constrained local correction and is reported with its own response diagnostics. Without these restrictions, wind and plume parameters could fit sparse pollution sensors while weakening the physical and source-level interpretation of $\widehat{\mathbf{c}}$ and the reconstructed activities.

5. Identifiability Theory for Source Apportionment

We now characterize when source activity at a chosen temporal-basis resolution is determined by the projected observations. Throughout this section,

$$\begin{aligned}\widetilde{Y} &= \widetilde{H}_\Phi \mathbf{c} + \widetilde{E}, \\ \widetilde{H}_\Phi &= P_Q^\perp H_\Phi^{\mathrm{lag}} \in \mathbb{R}^{N\times J}, \\ J &= KB.\end{aligned} \tag{37}$$

The analysis is conditional on the constructed wind field, transport operator, source inventories, temporal basis, lag window, and background basis. We use the flattened coefficient index $j = (k, b)$; "source fingerprint" without a basis qualifier refers only to the constant-activity case $B = 1$.

5.1. Exact Identifiability

Definition 5.1 (Exact identifiability)**.** The source–basis coefficient vector is identifiable from $\widetilde{Y}$ if

$$\widetilde{H}_\Phi \mathbf{c}_1 = \widetilde{H}_\Phi \mathbf{c}_2 \quad \Longrightarrow \quad \mathbf{c}_1 = \mathbf{c}_2. \tag{38}$$

This is identifiability of the reconstructed source activity trajectories at the selected temporal-basis resolution.

Proposition 5.2 (Identifiability of inventory-based apportionment)**.** *In the noiseless projected model $\widetilde{Y} = \widetilde{H}_\Phi \mathbf{c}$, $\mathbf{c}$ is identifiable uniformly for all $\mathbf{c} \in \mathbb{R}_+^J$ if and only if*

$$\mathrm{rank}(\widetilde{H}_\Phi) = J. \tag{39}$$

Proof. If $\mathrm{rank}(\widetilde{H}_\Phi) = J$, then $\ker(\widetilde{H}_\Phi) = \{0\}$, so $\widetilde{H}_\Phi(\mathbf{c}_1 - \mathbf{c}_2) = 0$ implies $\mathbf{c}_1 = \mathbf{c}_2$, including on the nonnegative orthant. Conversely, if $\mathrm{rank}(\widetilde{H}_\Phi) < J$, there exists a nonzero $\Delta\mathbf{c} \in \ker(\widetilde{H}_\Phi)$. Choose an interior $\mathbf{c}$ with entries large enough that both $\mathbf{c}$ and $\mathbf{c} + \alpha\Delta\mathbf{c}$ are nonnegative for some nonzero, sufficiently small α. They have the same response, so uniform identifiability on $\mathbb{R}_+^J$ fails. $\square$

Nonnegativity can remove some null directions at the boundary of the feasible cone, so a particular boundary point may be locally more identifiable than this uniform condition indicates. We do not make a tangent-cone or fitted-active-set identifiability claim, and a near-zero estimate does not justify deleting its column after fitting. Coefficients fixed to zero by scientific knowledge are instead masked before fitting; the theorem and all diagnostics then apply to the reduced, predeclared column set. Rank deficiency may arise from spatial source ambiguity, dependence among temporal modes, or their interaction.

5.2. Noise-Robust Identifiability

Let $\sigma_1(\widetilde{H}_\Phi) \geq \cdots \geq \sigma_J(\widetilde{H}_\Phi) \geq 0$ be the square roots of the eigenvalues of $\widetilde{H}_\Phi^\top \widetilde{H}_\Phi$, including zeros when $J > N$ or the matrix is rank deficient. Even with full rank, small singular values amplify noise.

Proposition 5.3 (Noise-robust apportionment)**.** *Assume $\widetilde{Y} = \widetilde{H}_\Phi \mathbf{c} + \widetilde{E}$ and $\operatorname{rank}(\widetilde{H}_\Phi) = J$. Let $\widehat{\mathbf{c}}_{\mathrm{LS}}$ be the unconstrained least-squares estimate. Then*

$$\|\widehat{\mathbf{c}}_{\mathrm{LS}} - \mathbf{c}\|_2 \leq \frac{\|\widetilde{E}\|_2}{\sigma_J(\widetilde{H}_\Phi)}. \tag{40}$$

If $\widehat{\mathbf{c}}$ is the nonnegative least-squares estimate and $\mathbf{c} \geq 0$, then

$$\|\widehat{\mathbf{c}} - \mathbf{c}\|_2 \leq \frac{2\|\widetilde{E}\|_2}{\sigma_J(\widetilde{H}_\Phi)}. \tag{41}$$

Proof. For least squares, $\widehat{\mathbf{c}}_{\mathrm{LS}} - \mathbf{c} = \widetilde{H}_\Phi^\dagger \widetilde{E}$, giving $\|\widehat{\mathbf{c}}_{\mathrm{LS}} - \mathbf{c}\|_2 \leq \|\widetilde{H}_\Phi^\dagger\|_2 \|\widetilde{E}\|_2 = \|\widetilde{E}\|_2/\sigma_J(\widetilde{H}_\Phi)$. For nonnegative least squares, the true $\mathbf{c}$ is feasible, so optimality gives $\|\widetilde{Y} - \widetilde{H}_\Phi \widehat{\mathbf{c}}\|_2 \leq \|\widetilde{E}\|_2$. Hence $\|\widetilde{H}_\Phi(\widehat{\mathbf{c}} - \mathbf{c})\|_2 \leq 2\|\widetilde{E}\|_2$. The lower singular-value bound completes the proof. □

Thus $\sigma_J(\widetilde{H}_\Phi)$ measures worst-case robustness: source apportionment can be exactly identifiable but practically unstable when this quantity is small.

5.3. Identifiability Diagnostics

For floating-point computation, the default numerical rank tolerance is

$$\begin{aligned} \tau_{\mathrm{num}} &= \max(N, J)\, \epsilon_{\mathrm{mach}}\, \sigma_1, \\ r_{\mathrm{num}} &= \#\{i : \sigma_i > \tau_{\mathrm{num}}\}. \end{aligned} \tag{42}$$

The primary identifiability score includes rank-deficiency zeros:

$$\mathcal{I} = \sigma_J(\widetilde{H}_\Phi). \tag{43}$$

The condition number is

$$\kappa(\widetilde{H}_\Phi) = \begin{cases} \sigma_1/\sigma_J, & r_{\mathrm{num}} = J, \\ \infty, & r_{\mathrm{num}} < J. \end{cases} \tag{44}$$

The smallest numerically positive value may additionally be reported as $\sigma_{\min,+}$, but it is not the identifiability score and never turns a rank-deficient matrix into a stable one. For a separately chosen, noise-dependent threshold τ_σ, the effective rank is

$$r_{\mathrm{eff}}(\tau_\sigma) = \#\{i : \sigma_i(\widetilde{H}_\Phi) > \tau_\sigma\}. \tag{45}$$

The threshold τ_σ is recorded independently of τ_{num}. If $r_{\mathrm{eff}}(\tau_\sigma) < J$, the sensor network does not support reliable estimation of all J source–basis coefficients at that noise level.

For coefficient-specific diagnostics, let $\widetilde{\mathbf{h}}_j$, $j = (k, b)$, be a projected source–basis fingerprint. Its visibility and the weak set at threshold $\tau_v \geq 0$ are

$$v_j = \|\widetilde{\mathbf{h}}_j\|_2, \qquad \mathcal{W} = \{j : v_j \leq \tau_v\}. \tag{46}$$

A weak coefficient has little measurable effect after transport and background correction. For eligible indices $i, j \notin \mathcal{W}$, pairwise ambiguity is measured by projected fingerprint coherence,

$$\rho_{ij} = \frac{|\widetilde{\mathbf{h}}_i^\top \widetilde{\mathbf{h}}_j|}{\|\widetilde{\mathbf{h}}_i\|_2 \|\widetilde{\mathbf{h}}_j\|_2}. \tag{47}$$

Values near one indicate nearly proportional sensor-time signatures. The ambiguous-pair set is

$$\mathcal{A} = \{(i, j) : i, j \notin \mathcal{W},\ \rho_{ij} > \tau_\rho\}, \qquad \tau_\rho \in [0, 1]. \tag{48}$$

Coherence is undefined when either fingerprint is weak. Such entries are reported as “N/A”, excluded from coherence extrema, and never allowed to hide the corresponding weak-visibility flags.

To express the same scale-invariant geometry as a distance, we reactivate the ray distance for eligible, generally signed projected fingerprints:

$$d_{ij}^{\mathrm{ray}} = \min_{\alpha \in \mathbb{R}} \frac{\|\widetilde{\mathbf{h}}_i - \alpha \widetilde{\mathbf{h}}_j\|_2}{\|\widetilde{\mathbf{h}}_i\|_2}. \tag{49}$$

For nonweak fingerprints,

$$d_{ij}^{\mathrm{ray}} = \sqrt{1 - \rho_{ij}^2}. \tag{50}$$

Numerical implementations evaluate $\sqrt{\max(0, 1 - \rho_{ij}^2)}$. This is an unoriented linear-ray diagnostic, not a claim that projected fingerprints are nonnegative. High coherence and small ray distance encode the same pairwise geometry, so ray distance is a complementary report rather than an independent identifiability theorem. Like coherence, it is “N/A” for pairs involving $\mathcal{W}$ and excluded from distance extrema.

5.4. Background Absorption

Background correction can improve robustness by removing broad temporal variation, but it can also remove source-driven signal. Let $P_Q = QQ^\dagger$. For coefficient fingerprint

$j = (k, b)$, define the background absorption fraction

$$a_j = \frac{\|P_Q \mathbf{h}_j^{\mathrm{lag}}\|_2}{\|\mathbf{h}_j^{\mathrm{lag}}\|_2}. \tag{51}$$

If $\|\mathbf{h}_j^{\mathrm{lag}}\|_2$ is zero at numerical tolerance, a_j is undefined and reported as N/A; the coefficient remains explicitly weak. Machine-readable reports use JSON `null`, not NaN. If $a_j \approx 1$, most of the raw source–basis fingerprint lies in the background space, and the projected visibility v_j will be small even if the raw fingerprint is large. This is why identifiability must be assessed after projection.

5.5. Response-Matrix Perturbations

The constructed response matrix can be wrong for two scientifically different reasons. Transport error includes wind interpolation, transport approximation, and dispersion parameters; inventory error includes missing, misplaced, misclassified, or incorrectly scaled proxy sources. Write the projected response discrepancy as

$$\Delta_H = \Delta_{\mathrm{tr}} + \Delta_{\mathrm{inv}} + \Delta_{\mathrm{int}}, \tag{52}$$

where the final term allows transport–inventory interaction. Let $\widetilde{H}_\Phi^\star$ be the true projected response and $\widehat{\widetilde{H}}_\Phi$ the constructed one, with

$$\|\widehat{\widetilde{H}}_\Phi - \widetilde{H}_\Phi^\star\|_2 \le \delta_H. \tag{53}$$

Then

$$\widetilde{Y} = \widehat{\widetilde{H}}_\Phi \mathbf{c} + \left[\widetilde{E} + (\widetilde{H}_\Phi^\star - \widehat{\widetilde{H}}_\Phi)\mathbf{c}\right]. \tag{54}$$

Response error therefore acts like additional observation error.

Proposition 5.4 (Perturbation bound)**.** *Assume $\widehat{\widetilde{H}}_\Phi$ has full column rank and $\|\widehat{\widetilde{H}}_\Phi - \widetilde{H}_\Phi^\star\|_2 \le \delta_H$. The unconstrained least-squares estimate using $\widehat{\widetilde{H}}_\Phi$ satisfies*

$$\|\widehat{\mathbf{c}} - \mathbf{c}\|_2 \le \frac{\|\widetilde{E}\|_2 + \delta_H \|\mathbf{c}\|_2}{\sigma_J(\widehat{\widetilde{H}}_\Phi)}. \tag{55}$$

The same norm bound applies to all three discrepancy terms, but their interpretations are not interchangeable. Wind and dispersion ensembles quantify transport uncertainty and may support uncertainty intervals under the ensemble model. Alternative inventory maps, locations, labels, or scales are inventory robustness scenarios, not confidence intervals. They are reported separately, and source-specific attribution remains conditional on the supplied inventory. Either error type is most damaging when the projected response is already ill-conditioned.

Table 1. Identifiability diagnostics reported with source activity estimates.

Diagnostic	Interpretation	Action
$r_{\mathrm{num}}(\widetilde{H}_\Phi)$	exact separability	require rank J
$\sigma_J(\widetilde{H}_\Phi)$	noise robustness	flag zero/small values
$\kappa(\widetilde{H}_\Phi)$	error amplification	set ∞ if deficient
r_{eff}	noise-level resolution	reduce grouping
v_j	coefficient visibility	flag weak coefficients
a_j	background absorption	warn after projection
ρ_{ij}	pairwise ambiguity	merge coherent groups
d_{ij}^{ray}	ray separation	report N/A if weak

5.6. Identifiable Source Resolution

When coefficient fingerprints are weak or highly coherent, separate source estimates can be misleading. We define the source ambiguity graph by

$$\mathcal{E}_{\mathrm{src}} = \{(k, k') : \exists b, b' \quad ((k, b), (k', b')) \in \mathcal{A}\}. \tag{56}$$

Each edge retains the maximum eligible coefficient coherence, minimum ray distance, and the source–basis pair attaining the trigger. Deterministically ordered connected components $\mathcal{G} = \{G_1, \ldots, G_M\}$ are recommended report groups. Weak coefficients remain separate flags and do not create edges. Rank deficiency with no eligible edge produces a global unresolved warning rather than an invented merge.

These components are deterministic, conservative reporting units, not a claim of the globally finest identifiable partition. In particular, if edges join A to B and B to C, the component contains $\{A, B, C\}$ even when A and C are distinguishable as a pair. The edge table therefore retains both triggering pairs and their diagnostic values so that this transitive over-merging is visible.

The coefficient fit is not silently replaced by a grouped refit. For reporting, group activity and fitted sensor contribution are the sums

$$\widehat{\theta}_{G_a}(t) = \sum_{k \in G_a} \widehat{\theta}_k(t), \qquad \widehat{Y}_{G_a} = \sum_{k \in G_a} \sum_b \mathbf{h}_{kb}^{\mathrm{lag}} \widehat{c}_{kb}. \tag{57}$$

Matrix diagnostics remain coefficient level; these group summaries express the coarser claims supported by the fit without inventing a scalar source fingerprint.

6. Identifiability-Aware Apportionment Method

We combine the estimator from Section 4 with the diagnostics from Section 5. The result is an identifiability-aware source apportionment procedure, abbreviated IASA, that estimates source contributions only at the resolution supported by the projected response matrix.

6.1. Inputs and Outputs

IASA takes sparse pollution observations and their row mask, sensor locations X_{sens}, wind observations and masks, source inventories $S \in \mathbb{R}^{q\times K}$, a background basis Q, a maximum lag L, a nonnegative temporal basis $\Phi \in \mathbb{R}_+^{T\times B}$, and an open-boundary transport response G^∂. It also uses a numerical SVD tolerance τ_{num} and distinct thresholds τ_σ, τ_ρ, and τ_v for effective rank, pairwise ambiguity, and visibility.

The outputs are source–basis coefficient estimates, reconstructed source activities and contribution estimates at the identifiable grouping, fitted trajectories and residuals, uncertainty intervals, singular-value, coherence, and ray-distance diagnostics, low-visibility flags, and merge recommendations.

6.2. Procedure

The procedure is:

1. **Predeclare the model.** Fix the primary background basis, physical lag-candidate grid, temporal bases, inventory version, and any scientifically justified fixed-zero coefficient mask before inspecting fitted source activities.

2. **Prepare wind.** Convert meteorological WD/WS to transport vectors, preserve masks and observed values for audit, train or load the FieldFormer wind imputer, query the response grid to obtain $\widehat{\mathbf{w}}_t(\mathbf{x}_g)$, and convert physical velocity to grid displacement.

3. **Build and select lagged responses.** For every unmasked source–basis pair and candidate lag, compute

$$\mathbf{h}^{\text{lag}}_{t,kb} = \sum_{\ell\in\mathcal{L}_t} \phi_b(t-\ell) O G^\partial_{t,t-\ell}(\widehat{\mathbf{w}}_{t-\ell:t})\mathbf{s}_k.$$

   Select the smallest L satisfying Equation 11, independently of $\widehat{\mathbf{c}}$, and retain the full lag-sweep diagnostics.

4. **Align, select, and project.** Form H^{lag}_Φ, with source-major, basis-minor columns; apply the observed-$PM_{2.5}$ selection identically to Y, H, Q, and row metadata; then compute

$$\widetilde{Y} = P_Q^\perp Y, \qquad \widetilde{H}_\Phi = P_Q^\perp H^{\text{lag}}_\Phi.$$

5. **Fit source activities.** Solve for coefficients and reconstruct activities:

$$\widehat{\mathbf{c}} = \arg\min_{\mathbf{c}\geq 0} \|\widetilde{Y} - \widetilde{H}_\Phi \mathbf{c}\|_2^2 + \lambda R(\mathbf{c}),$$
$$\widehat{\theta}_k(t) = \sum_b \widehat{c}_{kb}\phi_b(t).$$

6. **Audit identifiability.** Compute r_{num}, the singular values, $r_{\text{eff}}(\tau_\sigma)$, coefficient visibility v_j, background absorption a_j, coherence ρ_{ij}, and ray distance d^{ray}_{ij}.

7. **Flag unsupported groups.** Mark weak coefficient fingerprints $\mathcal{W} = \{j : v_j \leq \tau_v\}$. For eligible pairs only, set $\mathcal{A} = \{(i,j) : i,j \notin \mathcal{W}, \rho_{ij} > \tau_\rho\}$.

8. **Recommend report groups.** Map eligible coefficient-pair warnings to source-graph edges and take deterministic connected components. Weak coefficients are flagged but do not create coherence edges. Group activities and fitted sensor contributions are sums of their members; the fine-resolution coefficient fit is not silently replaced by a grouped refit.

9. **Check model adequacy.** When an externally calibrated observation-noise model is available, run the refitted parametric-bootstrap residual test below. Otherwise label the residual summary uncalibrated.

10. **Report.** Return source estimates, observation/transport uncertainty, inventory robustness scenarios, residual adequacy, diagnostics, and conservative source groups.

This procedure separates two questions: which source activity trajectories best fit the projected sensor data, and which source–basis coefficients are supported by the geometry of $\widetilde{H}_\Phi$.

The nonnegative quadratic problem is solved with projected FISTA in PyTorch. Each gradient step is projected by $\mathbf{c} \leftarrow \max(\mathbf{c}, 0)$, with a Lipschitz step derived from the largest singular value, monotone restart, and convergence checks based on the projected-gradient/KKT residual and relative objective change. Device, dtype, convergence status, iteration count, and the final KKT residual are part of the fit record. Transport-ensemble and bootstrap refits are batched on the same device when their shapes agree.

6.3. Uncertainty Reporting

Let

$$\widetilde{r} = \widetilde{Y} - \widetilde{H}_\Phi \widehat{\mathbf{c}} \tag{58}$$

be the projected residual. A residual variance estimate is

$$\widehat{\sigma}^2 = \frac{\|\widetilde{r}\|_2^2}{\max(N - r_{\text{eff}}(\tau_\sigma), 1)}. \tag{59}$$

For an unconstrained or active-set approximation, let $\mathcal{A}_c = \{j : \widehat{c}_j > \tau_c\}$ and let $\widetilde{H}_{\Phi,\mathcal{A}_c}$ contain the active columns. The covariance of the active estimates is approximated by

$$\text{Cov}(\widehat{\mathbf{c}}_{\mathcal{A}_c}) = \widehat{\sigma}^2(\widetilde{H}^\top_{\Phi,\mathcal{A}_c}\widetilde{H}_{\Phi,\mathcal{A}_c} + \lambda I)^\dagger. \tag{60}$$

For each wind-field ensemble member from Section 4.1, IASA constructs $\widetilde{H}_{\Phi}^{(b)}$, refits $\widehat{\mathbf{c}}^{(b)}$, and reports empirical quantiles across ensemble members. These intervals capture uncertainty induced by wind interpolation and transport-response construction. Every ensemble is tagged by provenance. Wind and dispersion members have `ensemble_kind=transport` and may form transport-uncertainty intervals. Alternative inventory maps, source locations, labels, or scales have `ensemble_kind=inventory` and produce scenario-wise robustness tables. The two kinds are never pooled into one interval. Active-set intervals describe the fitted observation model and likewise do not convert inventory scenarios into probabilistic uncertainty.

6.4. Residual Model-Adequacy Check

Residual magnitude alone is not calibrated evidence of misspecification. Let Z be an orthonormal basis for the background-orthogonal observed-row space, and define the nondegenerate residual coordinates and externally supplied noise covariance

$$\bar{r} = Z^{\top}(Y - H_{\Phi}^{\mathrm{lag}}\widehat{\mathbf{c}}), \qquad \bar{\Sigma}_e = Z^{\top}\Sigma_e Z. \tag{61}$$

When $\bar{\Sigma}_e$ is calibrated independently of this fitted residual and is positive definite, the adequacy statistic is

$$T_{\mathrm{res}} = \bar{r}^{\top}\bar{\Sigma}_e^{-1}\bar{r}. \tag{62}$$

IASA draws $B_{\mathrm{res}} = 1000$ parametric-bootstrap observations from the fitted source/background model plus the declared noise model, repeats the full coefficient fit for every draw, and recomputes T_{res}. At $\alpha = 0.05$, it flags model inadequacy when the observed statistic exceeds the empirical $(1-\alpha)$-quantile; the Monte Carlo p-value uses the standard add-one correction.

If no externally calibrated noise model is supplied, IASA reports raw and projected residual norms and sensor-wise, time-wise, and autocorrelation summaries with `calibration_status=uncalibrated`; it emits no pass/fail adequacy claim. Rejection establishes a mismatch with the declared model but does not identify a missing source. Non-rejection does not establish inventory completeness: in particular, an omitted source whose sensor-time signature lies in $\mathrm{span}[H_{\Phi}^{\mathrm{lag}}, Q]$ can be absorbed by fitted terms and remain residual-invisible.

6.5. Wind-Distribution Diagnostics

Diagnostics for one realized wind window answer a retrospective question. Prospective network adequacy is assessed empirically over predeclared historical or simulated wind windows. Across those response matrices, IASA reports the 5th, 50th, and 95th percentiles of σ_J, the probability of full numerical and effective rank, the probability that each coefficient is weak, the probability that each source pair is ambiguous, and the frequency of each conservative report component. These are Monte Carlo or empirical distributional summaries under the sampled wind population, not a new identifiability theorem.

6.6. Acceptance Criteria for Refinement

If the future constrained refinement in Section 4.5 is used, it is accepted only if it improves fit without degrading separability. Let $\widetilde{H}_{\Phi,0}$ be the projected response before refinement and $\widetilde{H}_{\Phi,\mathrm{ref}}$ after refinement. We require

$$\sigma_J(\widetilde{H}_{\Phi,\mathrm{ref}}) \geq (1-\eta_{\mathrm{id}})\sigma_J(\widetilde{H}_{\Phi,0}) \tag{63}$$

and

$$\max_{\substack{i \neq j \\ i,j \notin \mathcal{W}}} \rho_{ij}^{\mathrm{ref}} \leq \tau_{\rho}^{\mathrm{ref}}. \tag{64}$$

These checks prevent end-to-end tuning from improving sensor fit by making source–basis fingerprints less distinguishable. If either response is numerically rank deficient, its σ_J is zero under the padded-spectrum convention of Section 5.3.

6.7. Reporting Protocol

For sources $k \neq k'$, an edge is added when at least one eligible coefficient pair $((k,b),(k',b'))$ belongs to $\mathcal{A}$. Each edge retains its maximum eligible coherence, minimum ray distance, and the basis pair attaining the trigger. Connected components are ordered by the original source indices and become conservative recommended report groups. An A–B–C edge chain therefore produces one component even if A and C are pairwise distinguishable; the edge report preserves the trigger for each link. Components are not asserted to be the globally finest identifiable partition. Rank deficiency without an eligible pair produces a global unresolved warning rather than an invented edge.

For each reported source group G_a, IASA reports

$$\begin{gathered} \widehat{\boldsymbol{\theta}}_{G_a,1:T}, \quad \mathrm{CI}(\widehat{\theta}_{G_a}), \quad \{v_{kb} : k \in G_a\}, \\ \max_{\substack{k \in G_a,\, k' \notin G_a \\ (k,b),(k',b') \notin \mathcal{W}}} \rho_{(k,b),(k',b')}. \end{gathered} \tag{65}$$

A source contribution is marked reliable only when its constituent reported coefficients are above the visibility threshold, its uncertainty interval is sufficiently narrow, and its eligible coherence with other reported groups is below the ambiguity threshold. Reports retain the basis names of every weak coefficient and the coefficient pair attaining each cross-source coherence maximum. Coherence and ray distance involving weak fingerprints are displayed as “N/A” and excluded from extrema; the weak flags remain visible.

Otherwise IASA labels the source as weakly visible, ambiguous with another source group, or merged into a coarser identifiable group.

7. New Delhi Case Study and Experimental Platform

The empirical study is anchored in New Delhi rather than in an abstract city. Its regulatory sensor geometry, pollution observations, wind observations, and source inventories all come from the same regional setting. This section defines that setting once. Section 8 then varies wind, noise, backgrounds, source ambiguity, activity bases, and forward-model error within this platform.

7.1. Government Observations and Sensor Network

The authoritative local table contains hourly government observations with columns `monitor_id`, `timestamp_round`, `AT`, `RH`, `WD`, `WS`, `pm10`, and `pm25` (CPCB, 2023; 2025). It spans 21,960 hourly timestamps from 1 May 2018 through 31 October 2020 in Indian Standard Time. After station processing the study has 32 output sensors. Approximately 74.9% of wind-direction entries, 72.8% of wind-speed entries, and 90.5% of $PM_{2.5}$ entries are observed. The saved data retain station identifiers, coordinates, timestamps, source column names, and a valid-value mask in which `True` means that the government record is present; an explicit missing mask stores its complement.

The two Pusa monitors are not treated as coincident independent sensors. `Pusa_IMD` and `Pusa_DPCC` are averaged by timestamp into `Pusa_averaged`; its coordinates are the mean of their coordinates and each variable is averaged over the available readings at that hour. This operation, rather than grid-cell deduplication, produces the final 32-sensor layout.

$PM_{2.5}$ is never imputed. Its flattened valid-value mask defines M_O in Equation 14; the same ordered sensor–time rows are retained in Y, H_{Φ}^{lag}, Q, and their metadata.

7.2. Wind Preparation

The observed WD/WS values are converted using Equation 18 before entering the FieldFormer wind imputation pipeline. The saved wind product retains raw station values, masks, station coordinates, held-out validation splits, model configuration, checkpoint metadata, and the gridded query field used by the response operator. The New Delhi runs use the FieldFormer-imputed grid wind field.

Table 2. Named New Delhi proxy inventories and temporal assumptions.

Inventory	Proxy activity assumption
Brick kilns	seeded alternating 12-hour blocks
Industries	mixed 24/7 and daytime operation
Population density	peaks near 07:00, 13:00, 19:00
Traffic 00	nearest traffic time slot
Traffic 06	nearest slot; zero map in crop
Traffic 12	nearest traffic time slot
Traffic 18	nearest traffic time slot

7.3. Inventories and Proxy Temporal Activities

The inventory loader returns seven named proxy maps in the fixed order shown in Table 2. Every 80×80 map is cropped by rows 21:61 and columns 16:56, giving a 40×40 study map, then divided by its own cropped 99th percentile. The maps are never summed into an aggregate source field. Brick-kiln and industry proxies follow the regional emissions inventory (Guttikunda & Calori, 2013), population density uses the gridded population product (Columbia University, 2018), and traffic intensity is derived from map data (Google, 2024).

Brick-kiln blocks use a recorded deterministic seed. The industry profile combines a continuous operating fraction with higher activity from 07:00 to 19:00. Population-related activity combines a baseline with smooth cooking peaks, while each traffic map is active in the nearest of the 00, 06, 12, and 18 hour slots. These are deterministic study assumptions, not observed emissions truth.

The `traffic_06` inventory is identically zero in the selected crop and therefore has a zero response column for every basis component. It is retained for provenance but must be reported as unsupported unless the map is replaced or excluded before fitting. Because every inventory uses independent proxy normalization, fitted coefficients and contributions are in normalized proxy units. They are not physical emission totals or physical source shares. No learned free residual source map is used; unexplained broad signal belongs to the background model or residual diagnostics.

7.4. Forward Models, Sensor Observations, and Backgrounds

Both forward paths use the New Delhi-derived 40×40 domain and regulatory sensor coordinates. The paper-facing response matrix uses the open-boundary Gaussian puff operator from Section 4.2, evaluates kernels directly at sensors, and starts from a zero-source, zero-initial-state baseline. This is the operator-matched generator for controlled H1–H4 experiments and the inference operator for observed $PM_{2.5}$.

For observed New Delhi runs, the initial pollution field is estimated from the first available regulatory observations using spatial kriging on the response grid. Its zero-source transported trajectory is subtracted from the sensor observations before fitting source activities. Source coefficients are therefore fitted to the residual sensor-time signal after removing the contribution of the initialized background field and the low-rank temporal background basis. Controlled operator-matched experiments may instead use a zero initial field when the synthetic data are generated from the same zero-initial-state response.

An auxiliary advection–diffusion simulator uses first-order upwind advection, a five-point Laplacian, a two-stage Heun update, diffusivity 3×10^{-4}, and edge-hold boundaries. It is not silently substituted for the open-boundary response. It is used only as a labeled H5 forward-model-mismatch generator. Likewise, kriged government initial conditions are confined to auxiliary PDE trials; their matched zero-source trajectory is subtracted before source fitting.

Normal backgrounds are built from the source-independent components in Section 4.3 with effective rank at most eight. The base controlled background has requested and effective rank four: a global constant, centered linear trend, and first local-clock daily sine and cosine. Evaluation additionally uses a redundant-column version and a labeled source-like stress basis.

7.5. Controlled and Observed-Data Study Modes

The controlled mode preserves New Delhi inventories and sensor geometry while assigning known nonnegative source–basis coefficients. It uses synthetic or real imputed wind, controlled observation noise, source splits, sensor-layout variants, and optional operator mismatch. Because its coefficients are known, it supports recovery and interval-coverage metrics.

The observed-data mode combines real $PM_{2.5}$, its observation mask, the imputed New Delhi wind field, the named normalized inventories, and the declared proxy temporal bases. It has no ground-truth source activities. Accordingly, it reports residuals, response geometry, uncertainty, weak and ambiguous coefficients, normalized proxy contributions, and recommended report groups rather than synthetic recovery error.

8. Evaluation

The evaluation uses the New Delhi platform in Section 7 and asks whether the geometry of $\widetilde{H}_\Phi$ predicts which coefficient, activity, and grouped-source claims are defensible. A shared base configuration is changed along one controlled axis at a time; this avoids conflating wind, source geometry, background flexibility, noise, and forward-model error.

8.1. Experiment Matrix

H1: conditioning predicts recovery. We vary source geometry and Gaussian observation noise at 0%, 1%, 5%, 10%, and 20% of the maximum clean sensor signal. For each trial we compare coefficient and reconstructed-activity errors with σ_J, numerical and effective rank, condition number, and visibility.

H2: coherent sources require grouped reporting. We form increasingly coherent source pairs through nearby spatial splits and matched transport paths. Individual-source errors are compared with errors in the summed activity and fitted sensor contribution of each recommended connected component. Edge reports retain the triggering source–basis pair.

H3: background correction can help or hurt. We compare no background, the normal rank-four basis from Section 7.4, a redundant-column basis with the same span, and a labeled source-like stress basis. The response geometry is audited through visibility and the removed/raw absorption ratio. The rank-four primary and every sensitivity variant are declared before fitting; no basis is selected from Y or from its source-recovery score.

H4: wind diversity and sensor geometry change resolution. Matched source, basis, and sensor sets are evaluated under constant, single-direction, diurnal, AR(1), multi-directional, and real imputed New Delhi wind. Sensor variants are the regulatory layout, seeded random layouts, downwind-focused layouts, and layouts selected to increase σ_J or reduce maximum eligible coherence. The same columns are required on both sides of every wind comparison. Separate historical and simulated window ensembles estimate the 5th, 50th, and 95th percentiles of σ_J, rank probabilities, coefficient weakness probabilities, pairwise ambiguity probabilities, and conservative component frequencies.

H5a: transport error is amplified by ill-conditioning. We perturb wind speed, wind direction, and dispersion, and separately generate observations with the auxiliary edge-hold PDE while fitting with the open-boundary puff response. The operator error norm, coefficient/activity error, residual, and singular spectrum are reported together. These tagged transport ensembles may support transport-uncertainty intervals.

H5b: inventory error changes the attribution target. We perturb inventory locations, spatial scales, category assignments, and map versions while holding transport fixed. Each alternative inventory is a separate robustness scenario. Its variation is not pooled with H5a and is not reported as a confidence interval.

Lag-window sensitivity. Physical travel and residence times define the candidate grid. We evaluate Equation 11, choose the smallest lag satisfying $\tau_L = 10^{-3}$, and report rank, conditioning, and report-component stability across the grid without using fitted source coefficients for selection.

Missing-source model adequacy. Controlled trials omit either a residual-visible source outside $\mathrm{span}[H_\Phi^{\mathrm{lag}}, Q]$ or an aligned source whose signature lies in that span. The refitted parametric-bootstrap test from Section 6.4 is evaluated for calibration and power. The aligned case demonstrates why non-rejection cannot certify inventory completeness.

Temporal-basis recovery. Known traffic diurnal, brick-kiln intermittent, industry day/night, and mixed source-basis coefficients are reconstructed at increasing noise. This study separates coefficient error from error in the reconstructed source activity trajectory.

Real imputed wind is used as a controlled transport input when coefficients are synthetically known. The observed-$\mathrm{PM}_{2.5}$ mode is a separate evaluation and is never assigned synthetic source-recovery metrics.

8.2. Metrics and Reporting

Controlled trials report absolute and relative coefficient error, reconstructed-activity error, per-source error, grouped activity and sensor contribution error, projected sensor residual, uncertainty-interval coverage, and correlations between diagnostic values and attribution error. All trials also report the padded singular spectrum, σ_J, numerical and effective rank, condition number, coefficient visibility, background absorption, eligible coherence and ray distance, weak flags, triggering coefficient pairs, and deterministic report groups. Definitions and undefined weak-pair conventions are those of Section 5. Lag convergence, the fixed-zero coefficient mask and index mapping, primary background specification, and ensemble provenance accompany these metrics. Transport uncertainty and inventory robustness occupy separate fields.

Observed New Delhi runs report raw and projected residuals, fitted sensor trajectories, normalized proxy coefficients and activities, sensor-space fitted contributions, observation/transport uncertainty intervals, inventory-scenario robustness, weak/ambiguous flags, conservative grouped contributions, and the residual calibration status. A reported percentage, if used, is explicitly a fraction of the fitted inventory-attributed sensor signal and not a physical-emission share.

8.3. H1 Results: Conditioning and Recovery

Results pending. The final table will contain noise level, source geometry, σ_J, numerical/effective rank, condition number, coefficient error, activity error, residual, and interval coverage; the accompanying figure will plot recovery error against σ_J and condition number.

8.4. H2 Results: Coherence and Grouped Reporting

Results pending. The table will report each source pair, triggering basis pair, eligible coherence, ray distance, individual errors, connected component, and grouped activity and sensor-contribution errors. Chain cases will retain every edge and show when connected components conservatively merge two sources that are pairwise distinguishable.

8.5. H3 Results: Background Stress

Results pending. The table will compare requested/effective background rank, dependent columns, coefficient visibility, absorption, singular values, recovery error, and residual for the no-background, normal, redundant, and source-like stress cases, together with each predeclared basis identifier.

8.6. H4 Results: Wind Diversity and Sensor Geometry

Results pending. The table will cross wind provider and sensor layout with σ_J, rank, effective rank, condition number, maximum eligible coherence, minimum ray distance, weak coefficients, report groups, and activity error. Wind comparisons will use identical source–basis columns. A second table will report the ensemble quantiles and rank, weakness, ambiguity, and component probabilities.

8.7. H5a Results: Transport Error

Results pending. The table will identify the true and fitted forward operators, perturbation magnitude, response error norm, σ_J, condition number, coefficient/activity error, projected residual, and transport-interval coverage.

8.8. H5b Results: Inventory Robustness

Results pending. The table will identify the primary and alternative inventory versions, location/scale/category perturbation, diagnostics, fitted activities, and grouped contributions. Rows are robustness scenarios, not confidence-interval draws.

8.9. Lag-Window Selection Results

Results pending. The table will report candidate lag, response convergence ratio, σ_J, numerical/effective rank, condition status, ambiguity components, selected lag, and the physical candidate-grid rationale.

8.10. Missing-Source Adequacy Results

Results pending. The table will report omitted-source alignment, signal strength, calibrated noise model, T_{res}, bootstrap quantile, p-value, rejection rate, and residual summaries. It will include both the residual-visible and span-aligned omissions.

8.11. Temporal-Basis Recovery Results

Results pending. The table will report source, basis name, true and estimated coefficients, coefficient error, activity-trajectory error, weak flags, and uncertainty coverage for traffic, brick kilns, and industries.

8.12. New Delhi Wind-Imputation Validation

Results pending. The table will report held-out station-time vector error, direction and speed error, mask coverage, query-grid resolution, checkpoint, seed, device/dtype metadata, and preservation of raw observations in the audit product. Because dense real wind truth is unavailable, gridded-field accuracy is additionally assessed in controlled synthetic wind experiments.

8.13. New Delhi Identifiability and Report Groups

Results pending. The table will report each coefficient's source and basis, visibility, absorption, weak flag, cross-source trigger pairs, singular spectrum summary, global warnings, and deterministic connected components.

8.14. New Delhi Proxy Apportionment and Uncertainty

Results pending. The table will report normalized proxy coefficients, reconstructed activities, fitted sensor-space contributions, active-set and transport-ensemble intervals, inventory-scenario robustness, and grouped summaries. The zero `traffic_06` inventory will be marked unsupported rather than assigned a substantive estimate.

8.15. New Delhi Sensor Fit and Residual Diagnostics

Results pending. The table and plots will report observed-row count, raw and projected residual norms, station/time residual summaries, fitted versus observed trajectories, residual autocorrelation, background-removed signal, response/wind provenance, and—when an external noise model is available— T_{res}, bootstrap quantile, p-value, and adequacy decision. Without that calibration the result will be labeled `uncalibrated` and will not be presented as a model-adequacy pass.

9. Discussion and Conclusion

This work frames sparse-sensor source apportionment as an identifiable-resolution problem. Known or proxy source inventories restrict the source space and make attribution meaningful, but they do not guarantee that source groups are separable from sparse observations. The separability question is governed by how inventory groups appear at the sensor network after wind-conditioned transport, finite lag, background correction, and noise.

The central object is the projected lagged source–basis response matrix

$$\widetilde{H}_\Phi = P_Q^\perp H_\Phi^{\mathrm{lag}} \in \mathbb{R}^{N\times J}, \qquad J = KB. \tag{66}$$

Exact identifiability at the selected temporal-basis resolution requires $\mathrm{rank}(\widetilde{H}_\Phi) = J$, while robust attribution depends on the singular values (including rank-deficiency zeros), effective rank, coefficient visibility, background absorption, pairwise coherence, ray distance, and perturbation sensitivity of $\widetilde{H}_\Phi$. The constant-activity source model is the special case $B = 1$. A model can fit sensor trajectories while still producing unstable or non-unique coefficient and activity estimates, so prediction accuracy alone is not evidence of identifiability at the claimed temporal resolution.

The practical implication is that source apportionment should report a conservative grouping supported by the sensing system. If two source groups have highly coherent projected fingerprints, the data may support their combined contribution but not their separate activities. Reporting them separately can create false precision. The proposed IASA protocol therefore estimates nonnegative source–basis coefficients, reconstructs source activity trajectories, audits the projected response matrix, flags weak coefficients and ambiguous source pairs, and recommends deterministic connected report groups when the fine-grained inventory resolution is not identifiable. Group activities and sensor contributions sum member estimates without silently replacing the coefficient fit. Each warning retains the temporal-basis pair that triggered it. Connected components are deterministic but need not be the globally finest identifiable partition: an ambiguity chain can conservatively group endpoints that are distinguishable from one another.

This perspective also changes how sensing systems and preprocessing choices should be designed. Adding sensors is useful when it improves visibility, conditioning, coherence, or ray separation of coefficient fingerprints, not merely when it increases spatial coverage. Wind diversity can expose source groups from multiple transport angles, while a single dominant wind regime can leave sources confounded. Background correction can remove broad temporal confounding, but a basis that is too flexible can absorb source-driven variation and reduce identifiability. Wind interpolation and transport-model errors act as additional observation error whose impact is amplified by ill-conditioning. Prospective claims about a network require diagnostics over a distribution of historical or simulated wind windows rather than one realized wind-field reconstruction.

The New Delhi platform makes these choices concrete: government $PM_{2.5}$ rows remain missing rather than being imputed, wind vectors are completed with a FieldFormer-based gridded imputer queried on the response grid, and independently normalized proxy inventories define relative rather than physical emission units. Controlled experiments and the observed-data study share this platform but require different claims: only controlled trials have source-activity ground truth.

The framework has several limitations. The response operator is an approximation to atmospheric transport and can be extended with richer meteorology, vertical mixing, deposition, chemistry, and boundary inflow. The theory is linear in source–basis coefficients conditional on fixed wind, transport, temporal basis, inventory, and background components; fully joint estimation introduces additional nonlinearities. The selected temporal basis also limits the activity patterns that can be recovered: finer temporal resolution is defensible only when the corresponding response columns remain identifiable. Finally, the framework assumes that relevant candidate source inventories are available. Missing source categories may be absorbed into available inventories or background terms, which should be treated as a model-misspecification risk rather than as identifiable attribution. A calibrated residual bootstrap can reject an inadequate declared model but cannot prove inventory completeness, especially when an omitted signature lies in the span of the fitted response and background. Transport ensembles can quantify uncertainty under their sampling model, whereas alternative inventories are separate robustness scenarios and must not be pooled into confidence intervals. The learned gridded wind field is still only as reliable as the sparse station network and its held-out validation; dense real wind truth is unavailable.

Despite these limitations, the identifiability view provides a concrete basis for more reliable policy-facing apportionment. It separates fitting the observed sensor signal from determining which source claims are supported by the sensing system. By using rank, singular values, visibility, background absorption, coherence, ray distance, lag and wind-distribution sensitivity, separated transport/inventory perturbations, and calibrated residual checks, source apportionment can report not only estimated contributions, but also the spatial and temporal resolution at which those conditional contributions can be trusted.

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References

Banks, H. T. and Kunisch, K. *Estimation techniques for distributed parameter systems*. Springer Science & Business Media, 2012.

Belis, C., Karagulian, F., Larsen, B. R., and Hopke, P. Critical review and meta-analysis of ambient particulate matter source apportionment using receptor models in europe. *Atmospheric Environment*, 69:94–108, 2013.

Belis, C., Favez, O., Mircea, M., Diapouli, E., Manousakas, M., Vratolis, S., Gilardoni, S., Paglione, M., Decesari, S., Mocnik, G., et al. European guide on air pollution source apportionment with receptor models—revised version 2019. *Publications Office, LU*, 2019.

Bhardwaj, A., Balashankar, A., Iyer, S., Soans, N., Sudarshan, A., Pande, R., and Subramanian, L. Comprehensive monitoring of air pollution hotspots using sparse sensor networks. *ACM Journal on Computing and Sustainable Societies*, 3(4):1–36, 2025a.

Bhardwaj, A., Balashankar, A., and Subramanian, L. Fieldformer: Physics-informed transformers for spatio-temporal field reconstruction from sparse sensors. *arXiv preprint arXiv:2510.03589*, 2025b.

Carrassi, A., Bocquet, M., Bertino, L., and Evensen, G. Data assimilation in the geosciences: An overview of methods, issues, and perspectives. *Wiley Interdisciplinary Reviews: Climate Change*, 9(5):e535, 2018.

Chen, C.-T. and Chen, C.-T. *Linear system theory and design*, volume 301. Holt, Rinehart and Winston New York, 1984.

Chen, L., Xu, J., Wu, B., and Huang, J. Group-aware graph neural network for nationwide city air quality forecasting. *ACM Transactions on Knowledge Discovery from Data*, 18(3):1–20, 2023.

Colton, D. L., Ewing, R. E., Rundell, W., et al. *Inverse problems in partial differential equations*, volume 42. Siam, 1990.

Columbia University, C. f. I. E. S. I. N. . C. Gridded population of the world, version 4 (gpwv4): Population density, revision 11, 20230129 2018. URL https://doi.org/10.7927/H49C6VHW.

CPCB. Air quality data. https://cpcb.nic.in/, 2023.

CPCB. Cpcb data portal. https://app.cpcbccr.com/ccr/#/caaqm-dashboard-all/caaqm-landing/caaqm-comparison-data, 2025.

Engl, H. W., Hanke, M., and Neubauer, A. *Regularization of inverse problems*, volume 375. Springer Science & Business Media, 1996.

Google. Google maps. https://maps.google.com, 2024. Accessed: 2026-03-30.

Guttikunda, S. K. and Calori, G. A gis based emissions inventory at 1 km × 1 km spatial resolution for air pollution analysis in delhi, india. *Atmospheric Environment*, 67:101–111, 2013. ISSN 1352-2310. doi: https://doi.org/10.1016/j.atmosenv.2012.10.040. URL https://www.sciencedirect.com/science/article/pii/S1352231012010229.

Hopke, P. K. Review of receptor modeling methods for source apportionment. *Journal of the Air & Waste Management Association*, 66(3):237–259, 2016.

Hopke, P. K., Dai, Q., Li, L., and Feng, Y. Global review of recent source apportionments for airborne particulate matter. *Science of The Total Environment*, 740:140091, 2020.

Iyer, S. R., Balashankar, A., Aeberhard, W. H., Bhattacharyya, S., Rusconi, G., Jose, L., Soans, N., Sudarshan, A., Pande, R., and Subramanian, L. Modeling fine-grained spatio-temporal pollution maps with low-cost sensors. *npj Climate and Atmospheric Science*, 5(1):76, 2022.

Karagulian, F., Belis, C. A., Dora, C. F. C., Prüss-Ustün, A. M., Bonjour, S., Adair-Rohani, H., and Amann, M. Contributions to cities' ambient particulate matter (pm): A systematic review of local source contributions at global level. *Atmospheric environment*, 120:475–483, 2015.

Li, Y., Yu, R., Shahabi, C., and Liu, Y. Diffusion convolutional recurrent neural network: Data-driven traffic forecasting. *arXiv preprint arXiv:1707.01926*, 2017.

Li, Z., Kovachki, N., Azizzadenesheli, K., Liu, B., Bhattacharya, K., Stuart, A., and Anandkumar, A. Fourier neural operator for parametric partial differential equations. *arXiv preprint arXiv:2010.08895*, 2020.

Li, Z., Zheng, H., Kovachki, N., Jin, D., Chen, H., Liu, B., Azizzadenesheli, K., and Anandkumar, A. Physics-informed neural operator for learning partial differential equations. *ACM/IMS Journal of Data Science*, 1(3):1–27, 2024.

Ljung, L. et al. Theory for the user. *System identification*, 1987.

Mircea, M., Calori, G., Pirovano, G., BELIS, C., et al. European guide on air pollution source apportionment for particulate matter with source oriented models and their combined use with receptor models. 2020.

Paatero, P. and Tapper, U. Positive matrix factorization: A non-negative factor model with optimal utilization of error estimates of data values. *Environmetrics*, 5(2):111–126, 1994.

Pant, P. and Harrison, R. M. Critical review of receptor modelling for particulate matter: a case study of india. *Atmospheric Environment*, 49:1–12, 2012.

Raissi, M., Perdikaris, P., and Karniadakis, G. E. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational physics*, 378:686–707, 2019.

Tarantola, A. *Inverse problem theory and methods for model parameter estimation*. SIAM, 2005.

Viana, M., Kuhlbusch, T. A., Querol, X., Alastuey, A., Harrison, R. M., Hopke, P. K., Winiwarter, W., Vallius, M., Szidat, S., Prévôt, A. S., et al. Source apportionment of particulate matter in europe: a review of methods and results. *Journal of aerosol science*, 39(10):827–849, 2008.

Wang, B., Sun, Z., Jiang, X., Zeng, J., and Liu, R. Kalman filter and its application in data assimilation. *Atmosphere*, 14(8):1319, 2023.

Watson, J. G., Zhu, T., Chow, J. C., Engelbrecht, J., Fujita, E. M., and Wilson, W. E. Receptor modeling application framework for particle source apportionment. *Chemosphere*, 49(9):1093–1136, 2002.

Wu, Z., Pan, S., Long, G., Jiang, J., and Zhang, C. Graph wavenet for deep spatial-temporal graph modeling. *arXiv preprint arXiv:1906.00121*, 2019.

Yu, B., Yin, H., and Zhu, Z. Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting. *arXiv preprint arXiv:1709.04875*, 2017.

A. Temporal-Basis Notation and Diagnostic Conventions

Table 3. Indices and objects in the temporal-basis apportionment model.

Symbol	Shape or range	Meaning
k	$1, \dots, K$	source-group index
b	$1, \dots, B$	temporal-basis index
$j = (k, b)$	$1, \dots, J$	source-major, basis-minor coefficient index
Φ	$T \times B$	nonnegative source-activity basis
C	$K \times B$	nonnegative source–basis coefficients
$\mathbf{c}$	$J = KB$	source-major vectorization of C
H_Φ^{lag}	$N \times J$	observed-row coefficient fingerprints
$\widetilde{H}_\Phi$	$N \times J$	background-projected fingerprints
$\widetilde{\mathbf{h}}_{kb}$	N	fingerprint for coefficient c_{kb}

The constant-activity model uses $B = 1$, $\phi_1(t) = 1$, and $c_{k1} = \theta_k$. In the general model, diagnostics are computed for the J coefficient fingerprints. Source-level activity trajectories are reconstructed as

$$\widehat{\theta}_k(t) = \sum_{b=1}^{B} \widehat{c}_{kb} \phi_b(t);$$

fingerprint columns are not summed to create an artificial source-level diagnostic vector.

For a source-level ambiguity summary, IASA reports the largest eligible coherence between coefficients belonging to different sources and retains the source–basis pair that attains it. Coherence and ray distance are undefined for pairs involving a coefficient with $v_j \le \tau_v$; tables display these entries as N/A, exclude them from pairwise extrema, and report the weak coefficient separately.

B. Implementation Invariants

The response is constructed first on the complete time-major, sensor-minor row index. The $PM_{2.5}$ observation mask then selects the same ordered rows from Y, H_Φ^{lag}, Q, and metadata. Source–basis columns are always source-major and basis-minor. Any row or column metadata mismatch is a hard error.

Paper-facing responses use integer source-cell centers, half-cell domain extents, fractional final advection substeps, a positive minimum dispersion age, direct sensor evaluation, finite lag, and complete removal after center exit. Kernel boundary/truncation loss and whole-release exit loss are recorded separately. Retained and dropped kernel mass sum to emitted kernel mass within the recorded quadrature tolerance; sensor observations are never summed as a mass proxy. The default puff baseline has zero source and zero initial state.

Normal backgrounds have effective rank at most eight and do not use inventory or response columns. Thin-SVD projection stores the numerical tolerance, singular values, retained U_r, dependent input columns, removed components, and orthogonality/idempotence residuals. A source-like column is permitted only when the run is labeled as a stress test.

The primary Q, lag-candidate grid, and fixed-zero coefficient set are declared before fitting. A fixed-zero mask removes columns before diagnostics and fitting and stores both reduced-to-original and original-to-reduced index maps. Fitted near-zero coefficients never modify the mask. Adjacent lag responses use identical rows and columns; the default convergence threshold is $\tau_L = 10^{-3}$.

C. Sanity Gates

Before paper experiments, the implementation must pass six small deterministic gates built from public APIs:

1. **S1—response.** On a 16×16 grid with four compact sources, four sensors, impulse/constant bases, and seed 123, verify response shape, downwind arrival, open-boundary exit, no wraparound, dispersion anisotropy, mass accounting, metadata, and exact repeatability.

2. **S2—projection.** Use the S1 response with a rank-four normal background to verify exact ordering, background removal, orthogonality, idempotence, redundant-column invariance, and the labeled source-like stress behavior.

3. **S3—diagnostics.** Verify orthogonal, duplicate, and weak matrices, including padded zeros, σ_J, infinite deficient condition status, weak-pair N/A values, and a matched multi-source eastward versus two-direction wind comparison.

4. **S4—fit.** Recover known nonnegative coefficients in noiseless/noisy well-conditioned cases, verify projected-FISTA KKT convergence, preserve exact nonnegativity, and recover the duplicate-pair sum when the individual split is nonunique.

5. **S5—report groups.** Verify the duplicate-source connected component, an A–B–C chain with both trigger edges retained, separated-source non-edge, independent weak flag, conservative-group flag, and summed group contribution.

6. **S6—end to end.** Build sources, wind, response, background, masked observations, diagnostics, fit, uncertainty, and report groups with one command and write a compact provenance-complete summary.

Task 10 experiments begin only after S1–S6 and the PyTorch CPU/CUDA parity checks pass.

D. PyTorch Solver and Uncertainty Protocol

After CSV/NPZ ingestion, numerical computation uses PyTorch tensors on an explicit device and dtype. The nonnegative coefficient objective is

$$f(\mathbf{c}) = \|\widetilde{Y} - \widetilde{H}_\Phi \mathbf{c}\|_2^2 + \lambda\|\mathbf{c} - \mathbf{c}_0\|_2^2, \qquad \mathbf{c} \geq 0. \tag{67}$$

Projected FISTA uses a Lipschitz step derived from σ_1 or a recorded power-iteration estimate, clamps every iterate to the nonnegative orthant, and restarts acceleration whenever the objective increases. Termination requires the configured projected-gradient/KKT residual and relative objective-change tolerances, or reports that the iteration cap was reached. The fit artifact stores convergence status, iteration count, final KKT residual, objective summary, device, and dtype.

Active-set covariance uses PyTorch linear algebra on $\widetilde{H}_{\Phi,\mathcal{A}}^\top \widetilde{H}_{\Phi,\mathcal{A}} + \lambda I$. Wind-ensemble and bootstrap refits are batched on the same device when shapes permit. Undefined weak-pair metrics are serialized as JSON `null`, never NaN or a false numerical distance.

Uncertainty artifacts distinguish `ensemble_kind=transport` from `ensemble_kind=inventory`. Only the former may be summarized as an ensemble uncertainty interval; inventory members remain named robustness scenarios, and an aggregation request that mixes the two kinds is an error.

E. Adequacy and Ensemble Result Contracts

A residual-adequacy result stores `calibration_status`, T_{res}, bootstrap quantile, Monte Carlo p-value, α, replicate count, and noise-model provenance. Paper runs use $\alpha = 0.05$ and 1000 refitted replicates. If the externally calibrated noise covariance is absent or invalid, numerical test fields are null and the status is `uncalibrated`; no boolean pass is emitted. Sensor, time, and autocorrelation residual summaries are retained in either case.

A lag-selection result stores every candidate L, adjacent convergence ratio, selected L, τ_L, physical grid rationale, row count, and diagnostic/group summaries. A wind-distribution result stores the sampled-window provenance; 5th, 50th, and 95th percentiles of σ_J; full numerical and effective-rank probabilities; coefficient weakness probabilities; pairwise ambiguity probabilities; and conservative-component frequencies.

F. Reproducibility and Artifact Provenance

Every paper run records the container image, repository revision, PyTorch and CUDA versions, device model, dtype, deterministic-algorithm settings, random seeds, source files and crop, per-source scales, wind-imputer architecture, held-out wind split, query grid, gridded wind product, and ensemble provenance, inventory identifier/hash/version, observation mask, response and transport configuration, wind realization, dispersion settings, background columns and rank tolerance, lag protocol, fixed-zero mask and index mapping, diagnostic thresholds, ensemble kind, solver settings, noise-model provenance, and parent artifact hashes. Generated wind products, response matrices, checkpoints, and experiment outputs are written to generated-data or log paths and are not treated as immutable source files.

Controlled-run artifacts contain true and estimated coefficients, reconstructed activities, response/projection results, diagnostics, uncertainty, report groups, and recovery metrics. Observed New Delhi artifacts omit unavailable ground truth and instead contain observed-row counts, fitted trajectories, raw and projected residuals, normalized proxy activities/contributions, uncertainty, weak/ambiguous flags, trigger pairs, and group summaries. A percentage is emitted only with an explicit denominator identifying it as a fraction of fitted inventory-attributed sensor signal.